Name: _____ Period: ____

Catch-Up Guide: Sections 2.1–2.5

General Physics · Unit 2: Vectors and Projectile Motion

Welcome back! While you were out, the class covered five lessons on **vectors**: arrows with a size and a direction. Everything in the rest of the unit is built on them. When we launch projectiles in Section 2.6, you will split each motion into a sideways part and an up-and-down part, and that is a vector skill. This guide walks you through each lesson in order so you can catch up on your own.

What you need

- Your **Unit 2 workbook**. You'll use the reference sheet (pages 3–4) and Sections 2.1–2.5 (pages 7–16).
- The **slides** for Sections 2.1–2.5: the printed **Slide Companion** (two slides per page, each one labeled), or the slides on Google Classroom. Each section below tells you which slides to read.
- A **calculator in degree mode**. To test it, $\tan 45^\circ$ should give exactly 1. If you see 1.62, you are in radians. **Desmos** works too: open desmos.com/scientific, click the wrench, and choose Degrees.
- A **ruler and protractor** for Section 2.2.

How to work through each section

1. Read the **Big ideas** in this guide.
2. Open the slides and follow the **slide map**. When a slide says "Practice · on your own", try it on paper **before** you look at the solution slide.
3. Study the **worked example** in this guide. Cover the steps and try it yourself, then check.
4. Do the ● **Core** workbook problems. The answer key is with your teacher, so bring your work in to check it.
5. Do the **Check yourself** questions and compare with the answers on the last page. If you miss one, reread that section's Big ideas and its worked example.

Suggested plan

Session	Work on	About
1	Section 2.1 Scalars and Vectors	40 min
2	Section 2.2 Drawing and Adding Vectors	50 min
3	Section 2.3 Components: Breaking Vectors Apart	50 min
4	Section 2.4 Resultants from Components	45 min
5	Section 2.5 Reference Frames and Relative Velocity	45 min
6	Get Ready for Quiz 1 (workbook pages 17–18), then see your teacher	40 min

Two sessions a day is fine. Don't skip ahead: each section uses the one before it.

My progress

- ☐ Section 2.1: slides, worked example, Core problems, Check yourself
- ☐ Section 2.2: slides, worked example, Core problems, Check yourself
- ☐ Section 2.3: slides, worked example, Core problems, Check yourself
- ☐ Section 2.4: slides, worked example, Core problems, Check yourself
- ☐ Section 2.5: slides, worked example, Core problems, Check yourself
- ☐ Get Ready for Quiz 1 page done, and work checked with my teacher

Section 2.1 · Scalars and Vectors

Workbook pages 7–8 · Slides: Section 2.1 Launch: Dropped vs. Launched

What happened in class: a demo where one ball was dropped and another was launched sideways from the same height, then the vocabulary used for the rest of the unit.

Big ideas

- **The demo:** a ball dropped from rest and a ball launched sideways from the same height land at the **same time**. Moving sideways doesn't change how fast a ball falls. You'll use this in Section 2.6.
- A **scalar** is just a size, a number with a unit: distance, speed, time, mass, temperature.
- A **vector** has a size (its **magnitude**) **and** a direction: displacement (change in position), velocity, acceleration, force. A good test is to ask whether it points somewhere.
- **Distance** counts every step. **Displacement** compares only where you started and where you ended: $\Delta x = x_f - x_i$. Displacement can be negative or zero. Distance can't be negative.
- **Signs show direction.** Right, east and up are positive. Left, west and down are negative.
- Average speed $= \dfrac{\text{distance}}{\text{time}}$ (a scalar). Average velocity $= \dfrac{\Delta x}{\Delta t}$ (a vector, so it keeps its sign).
- Two vectors added together give a **resultant**. Its size is always between their difference and their sum. For example, 3 m and 5 m add to anything from 2 m to 8 m.

Follow the slides

Slides	What to do
1–3	Today's goals and the warm-up. This is Unit 1 review, so try it, then check slide 3.
4–8	The dropped vs. launched demo. Predict on slide 6 before you read slide 7. Slide 8 is a preview; just get the idea.
9–13	Scalars and vectors. Do the vocabulary sort on slide 11 before you look at the answers on slide 12.
14–18	Distance vs. displacement on a number line, the Δ symbol, average speed and velocity. Copy slide 14 into your notes.
19–23	Describing vectors, the sign rules, how big a sum can be, and three words: projectile, trajectory, range.
24–27	Practice 1 and 2. Try each on paper, then check the solution slide.
28–30	Model problem with its scoring guide. This shows what a full-credit quiz answer looks like.
31–32	Cheat sheet and exit ticket. Do the exit ticket as your last check.

Worked example

You walk 45 m east down the hallway to the gym, then 15 m west back to a water fountain. The walk takes 40 s. Take east as positive and start at $x_i = 0$. Find the distance, the displacement, the average speed, and the average velocity.

1. **Sketch it on a number line.** You start at 0, go out to +45 m, then come back to +30 m. So $x_f = +30$ m.
2. **Distance** adds every step and ignores direction: 45 m + 15 m = **60 m**.
3. **Displacement** compares the end with the start: $\Delta x = x_f - x_i = 30 - 0 =$ **+30 m** (30 m east). Signed legs give the same answer: $(+45) + (-15) = +30$ m.
4. **Average speed** $= \dfrac{60\text{ m}}{40\text{ s}} =$ **1.5 m/s**. It has no direction.
5. **Average velocity** $= \dfrac{+30\text{ m}}{40\text{ s}} =$ **+0.75 m/s** (0.75 m/s east).

Why the answers differ: you doubled back, so you covered more ground (60 m) than you ended up from the start (30 m). Average velocity can never be bigger than average speed.

Watch out

- A negative sign means a direction (west, down), not "less".
- Temperature can be negative, but it's still a scalar. It doesn't point anywhere.
- One full lap of a track is a big distance but **zero** displacement.

Your workbook problems

Workbook pages 7–8. Read the worked example at the top of the section first. Then do every ● Core problem. The hints below are only nudges, not answers.

Core	Hint
● 1	For each one, ask whether it points somewhere. A direction word makes it a vector.
● 2	Where does the runner end up compared with the start?
● 3	Follow the worked example step by step. Use 1.0 h for the time, so the answers come out in km/h.
● 5	Up is +12 and down is −20. Add them for Δy, and add their sizes for the distance.
● 9	West is negative, east is positive.
● 12	Use the word "direction" in your sentence.

○ **More practice:** 4, 6, 7, 8, 10. Do a few of these if a Core problem gave you trouble. ★ **Challenge:** 11, if the Core problems felt easy.

Check yourself (answers on the last page)

1. Label each S or V: (a) 15 m/s west (b) 3.0 kg (c) 9.8 m/s^2 down.
2. Take north as positive. You walk 20 m north, then 50 m south, in 35 s. Find the distance, the displacement Δy, the average speed, and the average velocity.
3. Could a 5 N pull and a 12 N pull add up to 4 N? What is the smallest and the largest they could add to?

Show your work here:

Questions for my teacher

Section 2.2 · Drawing and Adding Vectors

Workbook pages 9–10 · Slides: Section 2.2 Drawing and Adding Vectors

What happened in class: students drew vectors to scale, added them tip-to-tail, and did the Vector Walk lab outside. Workbook problem 1 is your version of the lab.

Big ideas

- **Drawing to scale:** choose a scale (for example 1 square = 5 m), draw each arrow with a ruler, and measure angles with a protractor.
- **Tip-to-tail:** draw the first arrow. Start the next arrow at the **tip** of the last one. The **resultant** goes from the tail of the first arrow to the tip of the last one. The order you add them in doesn't matter.
- **Naming a direction:** "37° N of E" means face east, then turn 37° toward north. An angle with no reference direction is not a direction.
- **Two perpendicular vectors** (like east and north) make a right triangle, so you can calculate instead of draw:

$$R = \sqrt{A^2 + B^2} \qquad \theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$$

- **Check your answer:** the resultant is longer than the longer leg and shorter than the two legs added together.

Follow the slides

Slides	What to do
1–4	Goals, the warm-up (try it, then check slide 3), and the orienteering hook.
5–6	How to draw to scale, and how to name a direction (N of E vs. E of N). Read slide 6 carefully.
7–10	Tip-to-tail, why order doesn't matter, and a full scale drawing.
11–13	The shortcut: the Pythagorean theorem and inverse tangent. Slide 12 shows why it works, and slide 13 explains $\tan^{-1}$.
14–16	Worked example (Mrs. Gordon's run), checking and reversing an answer, and five common mistakes.
18	The Vector Walk lab. You missed it, so do workbook problem 1 on the grid instead.
19–22	Practice 1 and 2. Try each first, then check the solution slide.
23–27	Model problem and scoring, cheat sheet, exit ticket.

Worked example

Mr. Jantz's dog Ivy runs 21 m south, then 20 m east. Find Ivy's displacement (magnitude and direction) and the distance she ran.

1. **Sketch:** a 21 m arrow pointing down, then a 20 m arrow pointing right from its tip. Draw the resultant from the start to the finish. The legs meet at 90°.
2. **Magnitude:** $R = \sqrt{21^2 + 20^2} = \sqrt{841} =$ **29 m**.
3. **Direction:** put the angle at the start, between south and the resultant. The side across from it is 20 m (east) and the side next to it is 21 m (south): $\theta = \tan^{-1}(20 \div 21) = 43.6°$, so **43.6° E of S**.
4. **Check:** 29 m is between 21 m and 21 + 20 = 41 m. ✓
5. **Distance:** 21 m + 20 m = **41 m**. That's more than 29 m, because Ivy didn't run in a straight line.

The same arrow can also be named from east: $\tan^{-1}(21 \div 20) = 46.4°$ S of E. Both names are correct, as long as you say which one you mean.

Watch out

- 3 m + 4 m is **not** 7 m unless both point the same way.
- Always write a reference direction, like "43.6° E of S", not just "43.6°".
- Mark the angle on your sketch first, so you know which side is opposite and which is adjacent.

Your workbook problems

Workbook pages 9–10. Read the worked example at the top of the section first. Then do every ● Core problem. The hints below are only nudges, not answers.

Core	Hint
● 1	1 square = 5 m, so 40 m is 8 squares. Measure the resultant in squares, then multiply by 5.
● 2	Fill in the blanks. Your answer should be a familiar right triangle (3-4-5).
● 3	Sketch first. North and west are perpendicular. Name the direction, like "__° W of N".
● 4	The way back is the same size, pointing the opposite way. Turn both direction words around.
● 9	These point opposite ways, on one line. Add them as signed numbers.

○ **More practice:** 5, 6, 7. Do a few of these if a Core problem gave you trouble. ★ **Challenge:** 8, if the Core problems felt easy.

Check yourself (answers on the last page)

1. You walk 16 m east, then 12 m north. Find your displacement (magnitude and direction) and the distance you walked.
2. A classmate writes "36.9°" as a direction. What's missing?
3. Add 10 m north and 4 m south.

Show your work here:

Questions for my teacher

Section 2.3 · Components: Breaking Vectors Apart

Workbook pages 11–12 · Slides: Section 2.3 Resolving Vectors into Components

What happened in class: Section 2.2 put two pieces together into one vector. This lesson goes the other way: it breaks one vector into a sideways (x) piece and an up-down (y) piece. This is the most important skill in the unit, so take your time on it.

Big ideas

- Any vector is the hypotenuse of a right triangle. Its two legs are its **components**: A_x (horizontal) and A_y (vertical).
- **SOH-CAH-TOA.** For an angle θ measured from the horizontal ($+x$ axis), these equations are on your reference sheet:

$$A_x = A\cos\theta \qquad A_y = A\sin\theta$$

- The rule that always works: **the side touching the angle gets cosine**. If the angle is measured from north, the north side gets cosine.
- **Signs come from your sketch:** left and down components are negative.
- **Going backward** (from components back to one vector): $A = \sqrt{A_x^2 + A_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|A_y|}{|A_x|}\right)$. Then name the direction from your sketch. Don't trust the calculator's signs.

Follow the slides

Slides	What to do
1–6	Goals, warm-up, and the big idea: every vector has a horizontal part and a vertical part.
7–9	Right-triangle trig: naming the sides, solving for a side, and why components work. Copy slide 7.
10	Worked example 1: a free kick. Copy it into your notes.
11–14	Signs in each quadrant, an angle measured south of west, and an angle measured from north. Read these slowly.
16–18	Going backward from components, the calculator trap, and five common mistakes.
19–22	Practice 1 and 2. Try each first, then check the solution slide.
23–27	Model problem (Mr. Vincenzo's two shots) and scoring, cheat sheet, exit ticket.

Worked example

(a) A soccer ball is kicked at 30 m/s at 35° above the horizontal, to the right. Find v_{ix} and v_{iy}.

1. **Sketch:** an arrow up and to the right, with 35° marked at the horizontal.
2. The horizontal side touches the angle, so it gets cosine: $v_{ix} = 30\cos 35^\circ =$ **+24.6 m/s** (right, so positive).
3. $v_{iy} = 30\sin 35^\circ =$ **+17.2 m/s** (up, so positive).
4. **Check:** $\sqrt{24.6^2 + 17.2^2} = 30$ m/s ✓. The components never add straight up to 30.

(b) A hiker's displacement is 12 m at 25° W of N. Take east and north as positive.

1. Face north, then turn 25° toward west. The angle touches the **north** side, so north gets cosine.
2. $y = +12\cos 25^\circ =$ **+10.9 m** (north).
3. $x = -12\sin 25^\circ =$ **−5.07 m** (west, so negative).

(c) Going backward. A puck has $v_x = -9.0$ m/s and $v_y = -4.0$ m/s. Find its velocity.

1. Size: $v = \sqrt{9.0^2 + 4.0^2} = \sqrt{97} =$ **9.85 m/s**.
2. Angle, using sizes only: $\theta = \tan^{-1}(4.0 \div 9.0) = 24.0^\circ$.
3. Both components are negative (left and down), so it's **24.0° below the −x axis** (24.0° S of W).

Watch out

- x doesn't always go with cosine. Cosine goes with the side that touches the angle.
- Radian mode is the number one lost point. $\sin 30^\circ$ should be 0.5.
- Left and down components need a minus sign.

Your workbook problems

Workbook pages 11–12. Read the worked example at the top of the section first. Then do every ● Core problem. The hints below are only nudges, not answers.

Core	Hint
● 1	Fill in the blanks. Sketch the arrow so you can see which side touches the 30°.
● 2	Part (b) points down, so y is negative. Sketch it first.
● 3	Same as worked example (a).
● 5	Same as worked example (c). Down and to the right means "below the +x axis".
● 7	Try 0°, 45° and 90° on your calculator and watch each component.
● 12	When is x the side across from (opposite) the angle?

○ **More practice:** 4, 6, 8, 9, 10. Do a few of these if a Core problem gave you trouble. ★ **Challenge:** 11, if the Core problems felt easy.

Check yourself (answers on the last page)

1. Resolve 40 m/s at 60° above the horizontal (to the right) into v_x and v_y.
2. Resolve 25 m at 20° S of E into x and y components (east and north positive).
3. A vector has $x = +5.0$ m and $y = -12$ m. Find its magnitude and direction.

Show your work here:

Questions for my teacher

Section 2.4 · Resultants from Components

Workbook pages 13–14 · Slides: Section 2.4 Resultants from Components

What happened in class: trips with three or four legs, like a drone flying east, then north, then west. Instead of drawing them, you add them up in a **component table**.

Big ideas

- A leg pointing exactly east, west, north or south goes straight into one column. East and west go in the x column, north and south go in the y column, and the other column gets a 0.
- Use signs: east +, west −, north +, south −.
- Add each column to get R_x and R_y. Then use the moves from Sections 2.2 and 2.3:

$$R = \sqrt{R_x^2 + R_y^2} \qquad \theta = \tan^{-1}\left(\frac{|R_y|}{|R_x|}\right)$$

- The signs of R_x and R_y tell you the direction: + and + is N of E, − and + is N of W, − and − is S of W, + and − is S of E.
- **The way back** has the same size and the opposite direction (S of W becomes N of E).
- Adding every number in the trip gives the **distance**, not the displacement.

Follow the slides

Slides	What to do
1–5	Goals, warm-up, the delivery drone hook, and a prediction. Answer slide 5 before you go on.
6–8	The long way (cosine and sine for every leg), the shortcut (compass legs go straight into one column), and why adding columns works.
9	The component table in four moves. **Copy this slide.**
10–14	Worked example 1 (Mr. Jantz's walk), naming the direction, and worked example 2 (negative parts and the way back).
16–17	How big the resultant can be, and five common mistakes.
18–21	Practice 1 and 2. Try each first, then check the solution slide.
22–24	Model problem (Ivy's fetch runs) and its scoring.
25	Optional challenge: a leg that isn't along a compass direction. Skip it if you're short on time.
26–27	Cheat sheet and exit ticket.

Worked example

A robot drives 7 m north, 10 m west, 3 m south, then 2 m west. Find its displacement and the way back to the start.

Leg	x part (east +)	y part (north +)
7 m north	0	+7 m
10 m west	−10 m	0
3 m south	0	−3 m
2 m west	−2 m	0
Sum	$R_x = -12$ m	$R_y = +4$ m

1. $R = \sqrt{12^2 + 4^2} = \sqrt{160} =$ **12.6 m**.
2. $\theta = \tan^{-1}(4 \div 12) = 18.4°$. R_x is negative (west) and R_y is positive (north), so the displacement is **12.6 m at 18.4° N of W**.
3. **Way back:** same size, opposite direction: **12.6 m at 18.4° S of E**.
4. **Distance driven:** 7 + 10 + 3 + 2 = 22 m, which is much more than 12.6 m.

Watch out

- Don't forget the minus sign on west and south legs. Write the compass word next to each entry.
- A north leg goes in the y column, not the x column.
- $\tan^{-1}(R_x \div R_y)$ gives the angle from north. That's only correct if your answer says "E of N" or "W of N".

Your workbook problems

Workbook pages 13–14. Read the worked example at the top of the section first. Then do every ● Core problem. The hints below are only nudges, not answers.

Core	Hint
● 1	Fill in the table. Every leg is east or north, so all the numbers are positive.
● 2	Two legs go in each column. The south leg is negative.
● 4	Find the displacement first, then turn it around for the way back.
● 8	Section 2.2 already showed this. Does the order of the legs matter?
● 10	When can two sizes simply add?

○ **More practice:** 3, 5, 6, 7. Do a few of these if a Core problem gave you trouble. ★ **Challenge:** 9, if the Core problems felt easy.

Check yourself (answers on the last page)

1. You walk 5 m east, 8 m north, then 9 m west. Find your displacement (magnitude and direction).
2. How far did you walk in question 1?
3. Which way, and how far, would take you straight back to the start?

Show your work here:

Questions for my teacher

Section 2.5 · Reference Frames and Relative Velocity

Workbook pages 15–16 · Slides: Section 2.5 Relative Velocity: Boats and Rivers

What happened in class: boats crossing rivers, walkways, and planes in the wind. Picture a toy car driving straight across a sheet of paper while someone slides the paper sideways: that is a boat in a river.

Big ideas

- A velocity depends on **who is measuring it** (the reference frame). A boat's speed relative to the water is not its speed relative to the shore.
- Add velocities as vectors, chaining the frames: $\vec{v}_{\text{boat rel. ground}} = \vec{v}_{\text{boat rel. water}} + \vec{v}_{\text{water rel. ground}}$.
- **On one line** (a walkway or a headwind), add them as signed numbers: with the walkway, add the speeds; against it, subtract them.
- **Crossing a river with the boat pointed straight across:** the current pushes only downstream. It doesn't change the crossing time at all. The two directions are separate, like the x and y columns.
- Each velocity is constant ($a = 0$), so the kinematic equation $y_f = \frac{1}{2}at^2 + v_i t + y_i$ becomes $y_f = v_{\text{across}}\, t$ when you start at $y_i = 0$. That gives:

$$t = \frac{\text{width}}{v_{\text{across}}} \qquad \text{drift} = v_{\text{current}}\, t$$

- The speed relative to the ground is the Pythagorean theorem of the two perpendicular velocities.

Follow the slides

Slides	What to do
1–4	Goals, warm-up, and the MythBusters ball-cannon hook.
5–7	Reference frames, and the rule for chaining frames. Copy slide 6.
8–9	Predict: does the current slow the crossing? Answer it before you look at slide 9.
10–12	Worked example the long way (component table) and the shortcut. Copy slides 10–11.
13–15	Why the current never changes the crossing time, what happens when the current doubles, and a plane in a crosswind.
16	Extension (rain seen by a runner). Optional.
18	Five common mistakes.
19–22	Practice 1 and 2. Try each first, then check the solution slide.
23–25	Model problem (Mr. White's rowing practice) and scoring.
26	Extension, not on Quiz 1: aiming upstream. Optional.
27–28	Cheat sheet and exit ticket.

Worked example

A boat points straight across a 90 m wide river at 3.0 m/s relative to the water. The current flows downstream at 1.2 m/s. Find the crossing time, the downstream drift, and the boat's velocity relative to the shore.

1. **Time uses only the across column:** $90 = (3.0)\,t$, so $t = 90 \div 3.0 =$ **30 s**.
2. **Drift uses only the downstream column,** over the same time: $(1.2)(30) =$ **36 m downstream**.
3. **Velocity relative to the shore:** the two velocities are perpendicular, so $v = \sqrt{3.0^2 + 1.2^2} =$ **3.23 m/s**.
4. **Direction:** $\theta = \tan^{-1}(1.2 \div 3.0) =$ **21.8° downstream from straight across**.

One-line version: on an airport walkway moving at 0.8 m/s, you walk at 1.4 m/s relative to the walkway. Walking with it, you move at 1.4 + 0.8 = 2.2 m/s relative to the ground. Walking against it, you move at 1.4 − 0.8 = 0.6 m/s.

Watch out

- Don't divide the width by the diagonal speed (3.23 m/s). Only the across speed carries you across.
- A faster current means **more drift**, not a longer crossing.
- A headwind subtracts. Don't add speeds that point opposite ways.

Your workbook problems

Workbook pages 15–16. Read the worked example at the top of the section first. Then do every ● Core problem. The hints below are only nudges, not answers.

Core	Hint
● 1	Fill in the blanks, then compare with the worked example.
● 2	This one is on a single line: add for (a), subtract for (b).
● 3	Same three steps as the worked example.
● 4	Which column does the current affect?
● 5	Airspeed and wind are perpendicular, like a boat and a current. Name the direction ("__° E of N").
● 10	Which speed sets the crossing time?

○ **More practice:** 6, 7, 8. Do a few of these if a Core problem gave you trouble. ★ **Challenge:** 9, if the Core problems felt easy.

Check yourself (answers on the last page)

1. A boat points straight across a 40 m river at 2.0 m/s relative to the water. The current is 0.50 m/s. Find the crossing time, the drift, and the speed relative to the shore.
2. In question 1, the current doubles to 1.0 m/s. What happens to the time and the drift?
3. A plane points due north at 150 m/s (airspeed). A 20 m/s wind blows toward the east. Find its velocity relative to the ground.

Show your work here:

Questions for my teacher

Getting Ready for Quiz 1

Quiz 1: Vectors and Relative Velocity · Workbook pages 17–18

Quiz 1 covers everything in this guide. The quiz uses perpendicular vectors and compass directions only, and you get the reference sheet (workbook pages 3–4).

Before the quiz

- ☐ Rate yourself on each Unit 2 learning target in the tracker at the back of the workbook.
- ☐ Do the ● Core problems on the Get Ready for Quiz 1 page: 1–7 and 12.
- ☐ Redo any Check yourself question you missed in this guide.
- ☐ Bring your workbook in so your teacher can check your Core problems with you.
- ☐ Check that your calculator is in degree mode.

Words to know

Word	Meaning
Scalar	A size only (distance, speed, time).
Vector	A size and a direction (displacement, velocity, acceleration, force).
Magnitude	The size of a vector, with no direction.
Displacement	Change in position: $\Delta x = x_f - x_i$.
Resultant	The single vector you get by adding two or more vectors.
Component	The x or y part of a vector.
Reference frame	The place a velocity is measured from: the ground, the water, the train.
Projectile, trajectory, range	An object moving under gravity alone, its curved path, and how far it travels sideways.

The four moves for almost every quiz problem

1. **Sketch** every vector and mark the angle.
2. **Split** anything that isn't on an axis into x and y parts, with signs.
3. **Add** each column.
4. **Combine** with the Pythagorean theorem and $\tan^{-1}$, then **name the direction** in words.

Check Yourself Answers

Compare after you try. Each answer shows the key step.

Section 2.1

1. (a) V (b) S (c) V
2. Distance = 20 + 50 = 70 m. $\Delta y = (+20) + (-50) = -30$ m (30 m south). Average speed = 70 ÷ 35 = 2.0 m/s. Average velocity = −30 ÷ 35 = −0.857 m/s (0.857 m/s south).
3. No. The smallest is 12 − 5 = 7 N (opposite ways) and the largest is 12 + 5 = 17 N (same way).

Section 2.2

1. $R = \sqrt{16^2 + 12^2} = 20$ m. $\theta = \tan^{-1}(12 \div 16) = 36.9°$, so 20 m at 36.9° N of E. Distance = 28 m.
2. A reference direction, such as "N of E". Without one, the angle could point anywhere.
3. They point opposite ways on one line: (+10) + (−4) = +6, so 6 m north.

Section 2.3

1. $v_x = 40 \cos 60° = +20.0$ m/s. $v_y = 40 \sin 60° = +34.6$ m/s.
2. $x = +25 \cos 20° = +23.5$ m. $y = -25 \sin 20° = -8.55$ m (south, so negative).
3. $\sqrt{5.0^2 + 12^2} = 13$ m. $\tan^{-1}(12 \div 5.0) = 67.4°$. Right and down, so 13 m at 67.4° below the +x axis (67.4° S of E).

Section 2.4

1. $R_x = +5 + (-9) = -4$ m, $R_y = +8$ m. $R = \sqrt{4^2 + 8^2} = 8.94$ m. $\tan^{-1}(8 \div 4) = 63.4°$, so 8.94 m at 63.4° N of W.
2. 5 + 8 + 9 = 22 m.
3. 8.94 m at 63.4° S of E.

Section 2.5

1. $t = 40 \div 2.0 = 20$ s. Drift = (0.50)(20) = 10 m downstream. Speed = $\sqrt{2.0^2 + 0.50^2} = 2.06$ m/s.
2. The time stays 20 s, because the current doesn't push across. The drift doubles to 20 m.
3. $\sqrt{150^2 + 20^2} = 151$ m/s. $\tan^{-1}(20 \div 150) = 7.59°$, so 151 m/s at 7.59° E of N.